

THE

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Aly Tan, Diandian Shi

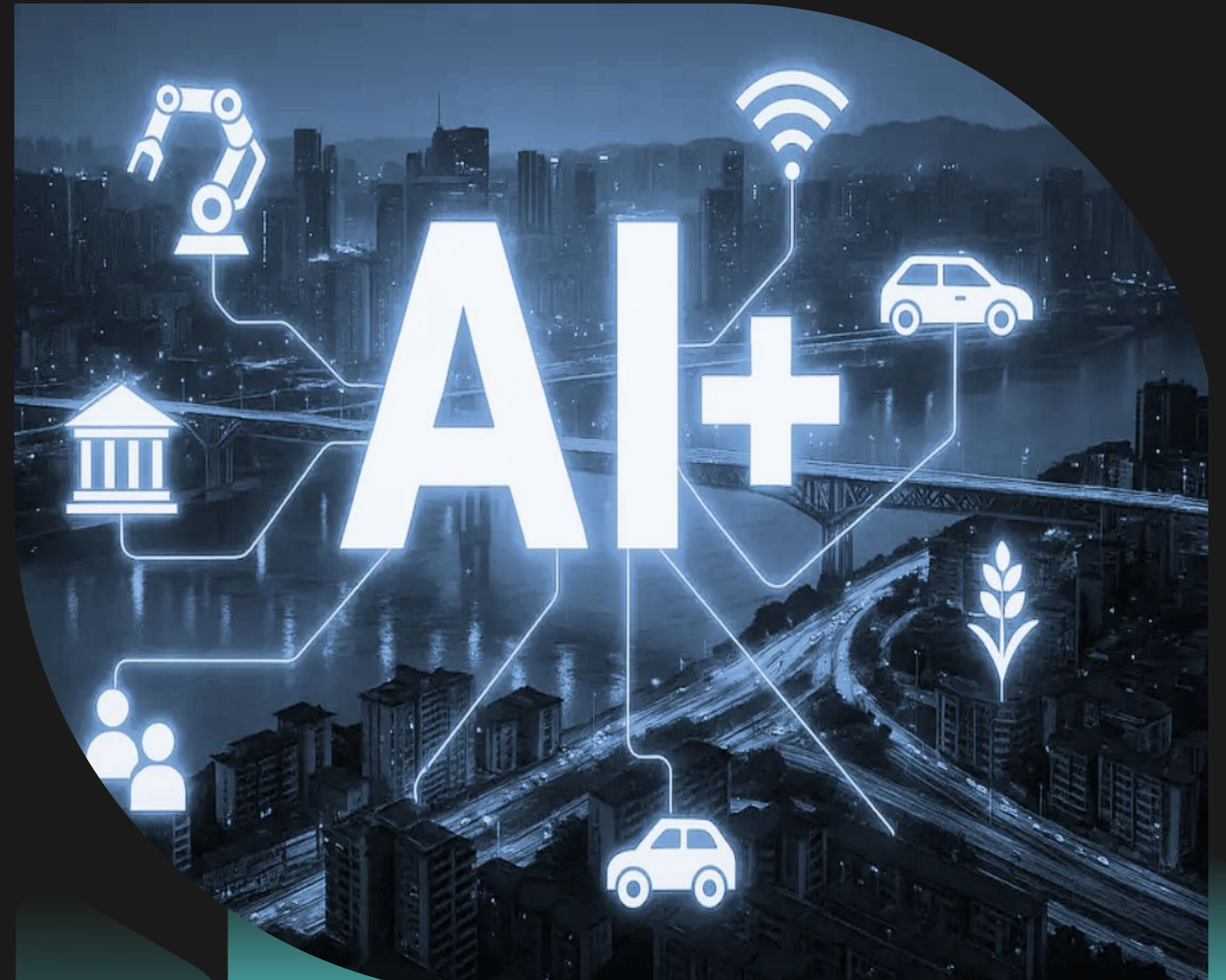
PACIFIC



DIVIDE

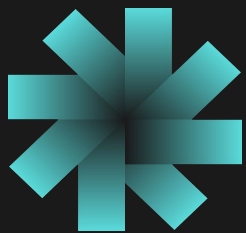
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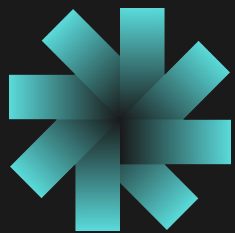
OBJECTIVES

subscriber data

01 

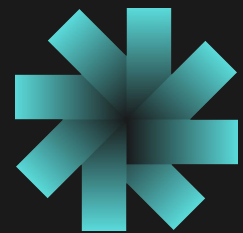
Which traffic
sources bring
high-quality
readers?

articles data

02 

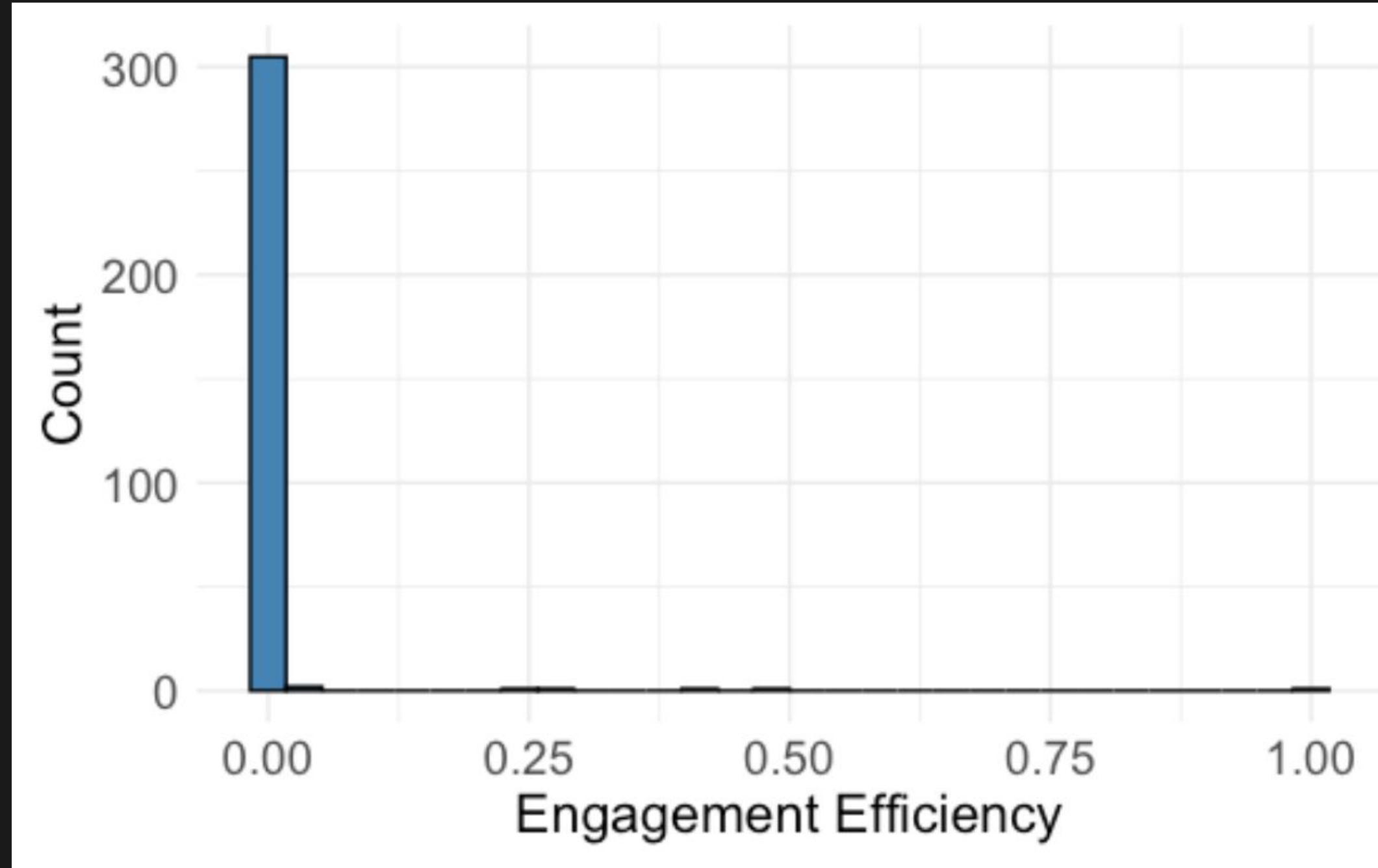
What title features
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drive audience
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measured by article
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articles data

03 

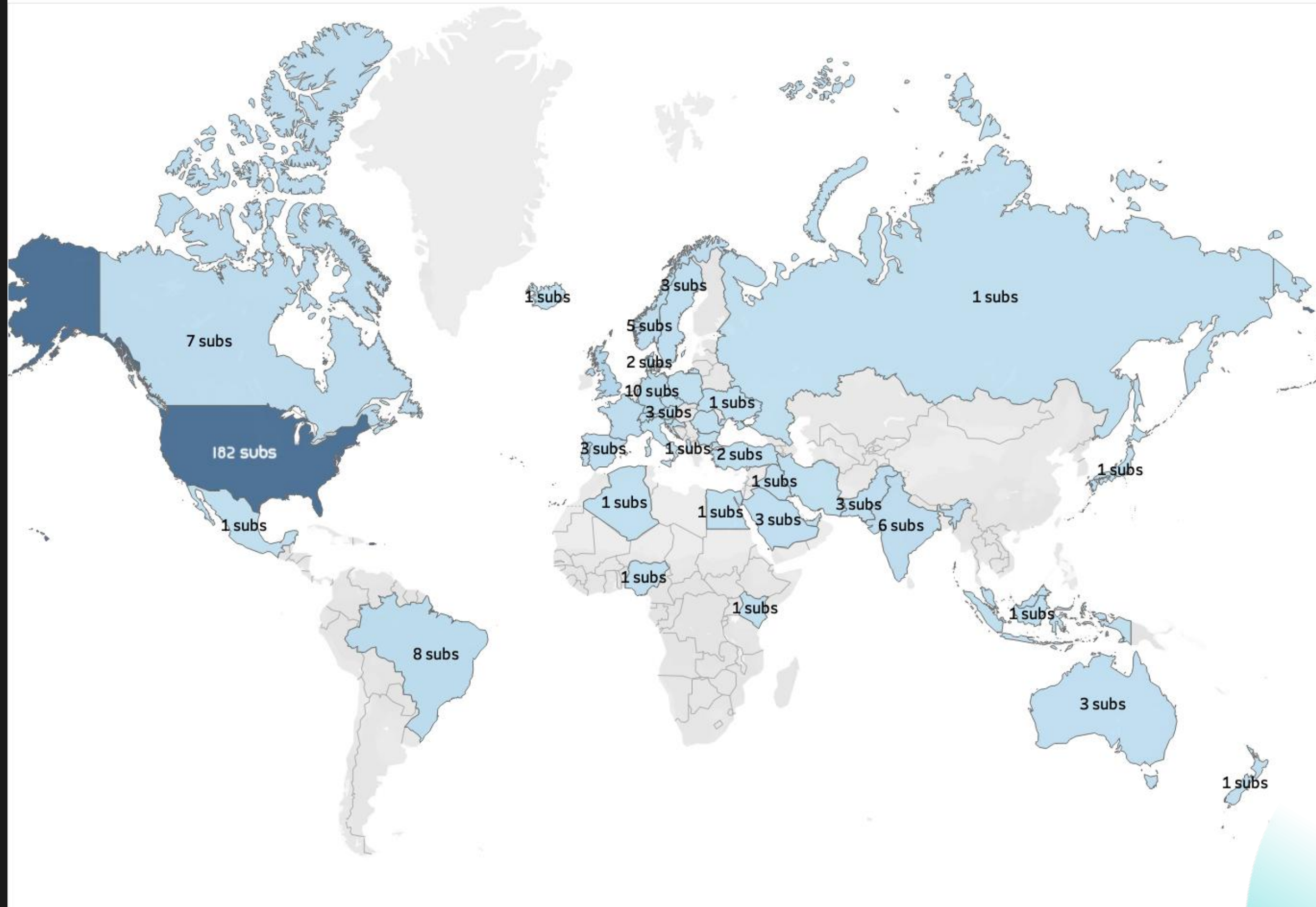
Which publishing day
and hour are
associated with
higher article views?

DESCRIPTIVE ANALYSIS



DESCRIPTIVE ANALYSIS

Top Subscriber Markets and Paid Performance



Summary:

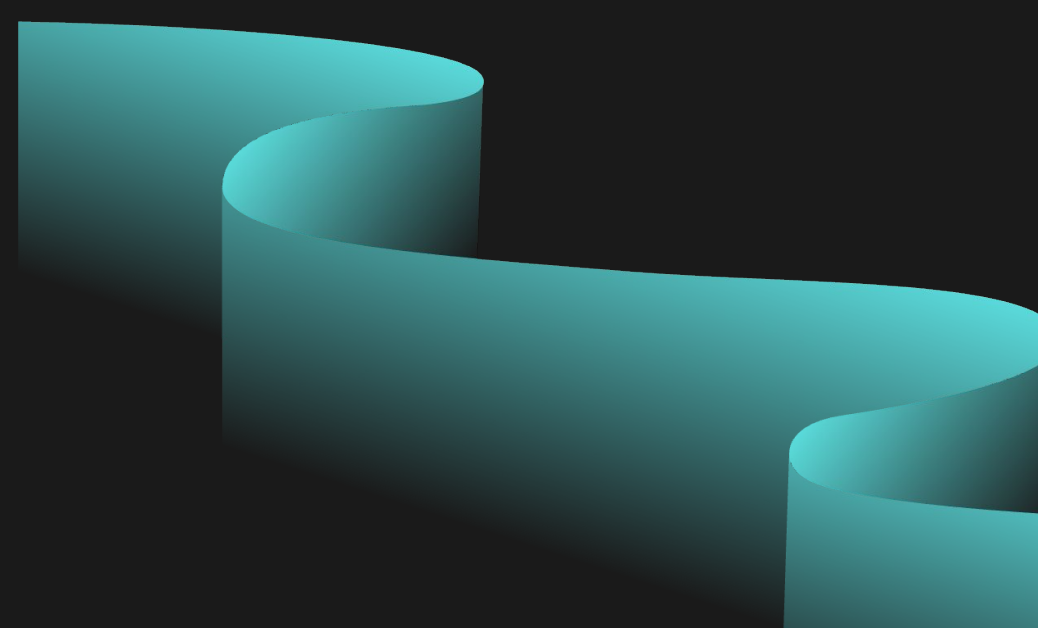
- 1) Audience is overwhelmingly composed of free subscribers, with only a small proportion converting to paid plans.
 - 2) Engagement is highly uneven. While many subscribers open emails and consume content, only a small subset actively interacts through commenting or sharing.
 - 3) Subscriber base is geographically concentrated in the United States, suggesting that audience growth may depend heavily on domestic reach.
- 
- Three overlapping teal circles of varying sizes are located in the bottom right corner of the slide, serving as a decorative element.

RESEARCH

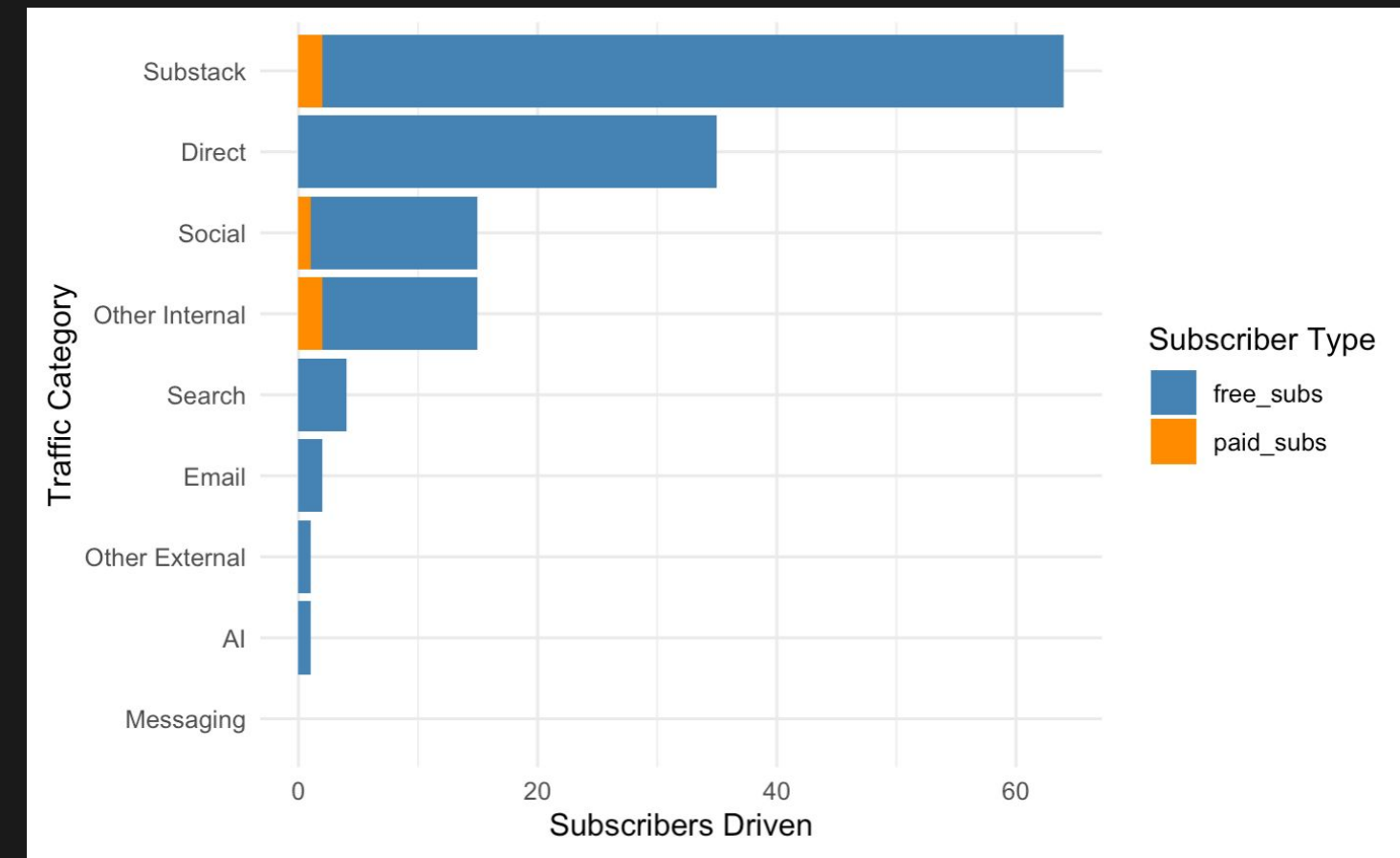
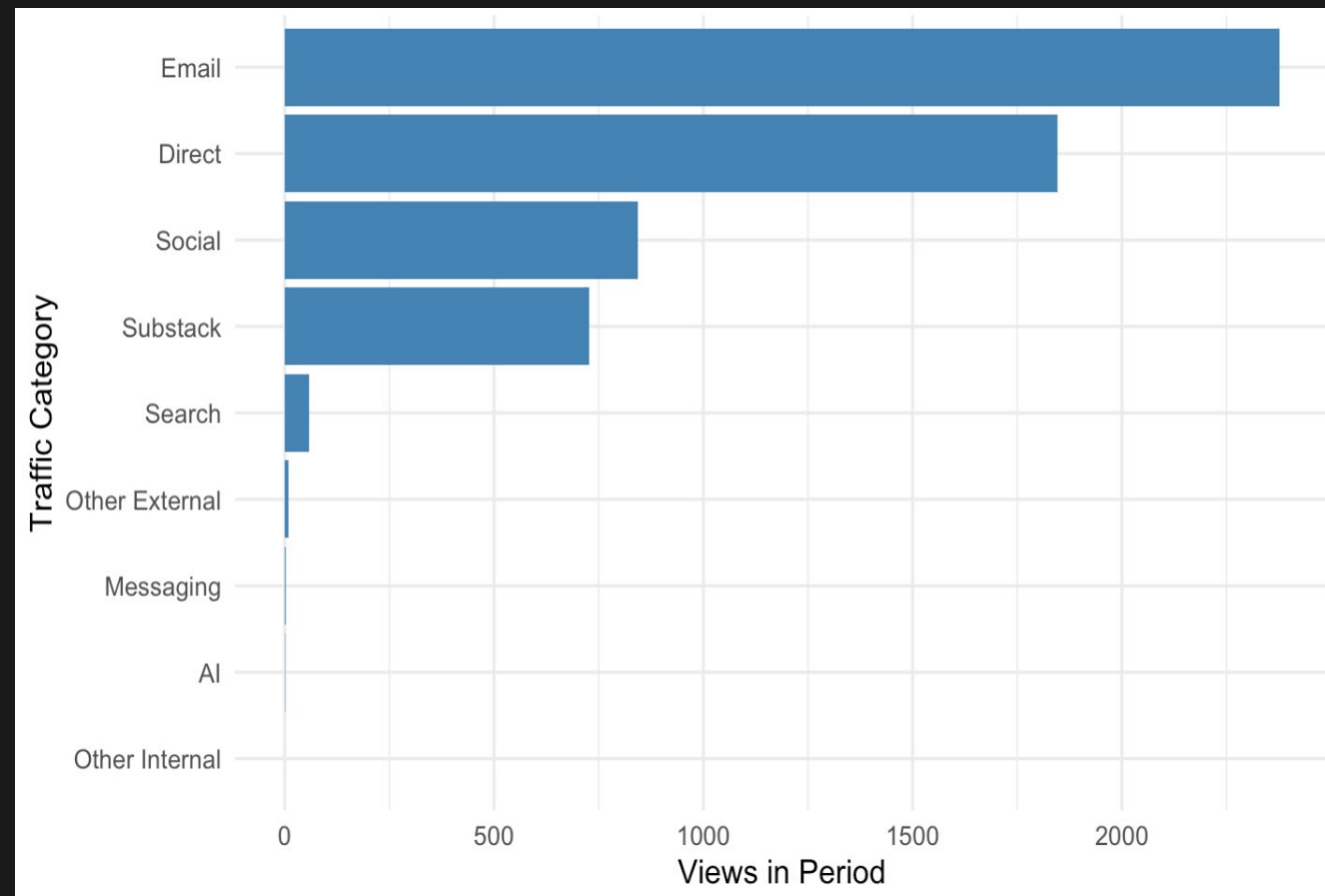
QUESTION 1 Which traffic sources bring high-quality readers?

High-quality readers: those who **view more unique posts**

METHODS:

- 1) recategorization of Subscription Source Free
 - 2) one-way ANOVA: whether the mean number of unique posts seen differs
 - 3) Tukey Post-Hoc Analysis: How it differs
- 

Exploratory Data Analysis (EDA): the traffic source dataset



- Email and Direct generate the **highest traffic volume**.
- **Substack drive the most subscribers**, especially free subscribers.

RESEARCH QUESTION 1

<u>Subscription Source</u>	<u>Free Count</u>
direct	36
substack-app	21
substack-reader-profile-discovery	20
linkedin.com	19
substack-notes	18
share	17
substack-signup-flow	14
post-page-subscribe-prompt	11
direct-to-app	10
post-end-cta	6
www.linkedin.com	5
m.facebook.com	4
import	3
substack.com	3
gift	2
l.facebook.com	2
www.google.com	2
bsky.app	1
com.reddit.frontpage	1
email-upsell	1
mail.google.com	1
open.substack.com	1
substack-follow-recommendations	1
substack-post-unlock	1
substack-your-weekly-stack	1
usc-word-edit.officeapps.live.com	1
lwww.noahpinion.blog	1

Grouped into broader
categories for analysis



Recategorized Traffic Sources:

- ❖ Direct
- ❖ Substack Network
- ❖ Substack Platform
- ❖ Social
- ❖ Article Page
- ❖ Shared Link
- ❖ Other

one-way ANOVA Results:

unique posts seen ~ traffic source

Source	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Traffic Source	6	339.2	56.53	3.871	0.00113**
Residuals	196	2862.3	14.60		

Statistically significant difference in the mean number of unique posts viewed across the different traffic sources.

Tukey's Post Hoc Analysis:

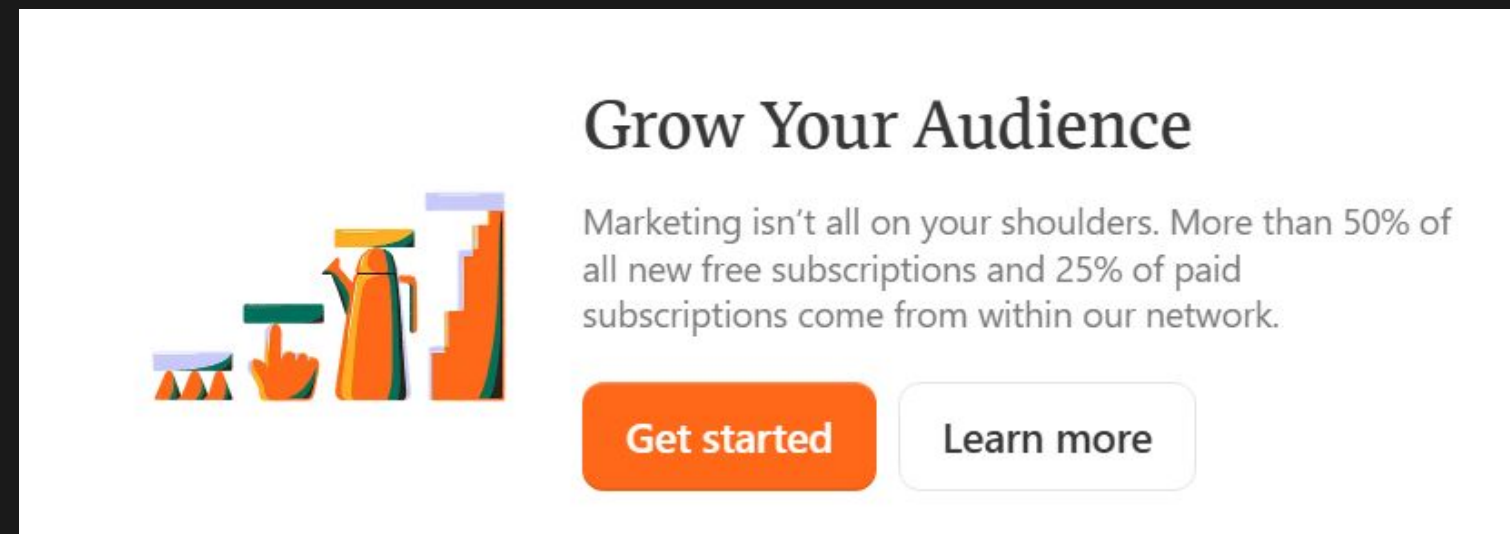
Comparison	Difference	Lower CI	Upper CI	Adj. p-value
Direct – Article Page	0.111	-3.054	3.276	0.9999
Other – Article Page	-0.389	-4.746	3.968	0.9999
Shared Link – Article Page	1.317	-2.533	5.167	0.9492
Social – Article Page	-0.295	-3.649	3.059	0.9999
Substack Network – Article Page	3.216	-0.041	6.474	0.0555
Substack Platform – Article Page	-0.047	-3.267	3.172	1.0000
Other – Direct	-0.500	-4.321	3.321	0.9997
Shared Link – Direct	1.206	-2.026	4.437	0.9239
Social – Direct	-0.406	-3.027	2.215	0.9993
Substack Network – Direct	3.105	0.610	5.601	0.0050
Substack Platform – Direct	-0.159	-2.604	2.287	1.0000
Shared Link – Other	1.706	-2.700	6.111	0.9103
Social – Other	0.094	-3.885	4.073	1.0000
Substack Network – Other	3.605	-0.293	7.503	0.0904
Substack Platform – Other	0.341	-3.524	4.207	0.9999
Social – Shared Link	-1.612	-5.029	1.805	0.7983
Substack Network – Shared Link	1.899	-1.423	5.221	0.6146
Substack Platform – Shared Link	-1.364	-4.649	1.920	0.8785
Substack Network – Social	3.512	0.780	6.243	0.0032
Substack Platform – Social	0.248	-2.438	2.933	0.9999
Substack Platform – Substack Network	-3.264	-5.827	-0.700	0.0037

Substack Network users view significantly **more posts** than users from Direct, Social and Substack Platform.

Most other pairs: **no significant difference** in engagement.

KEY TAKEAWAYS

Which traffic sources bring high-quality readers?



This was screenshotted directly off of the Substack Website. The statement correlates with our results.

Readers from the **Substack Network** show the highest engagement.

They view **significantly more unique posts** than readers from Direct or Social traffic.

Internal Substack discovery attracts higher-quality readers.

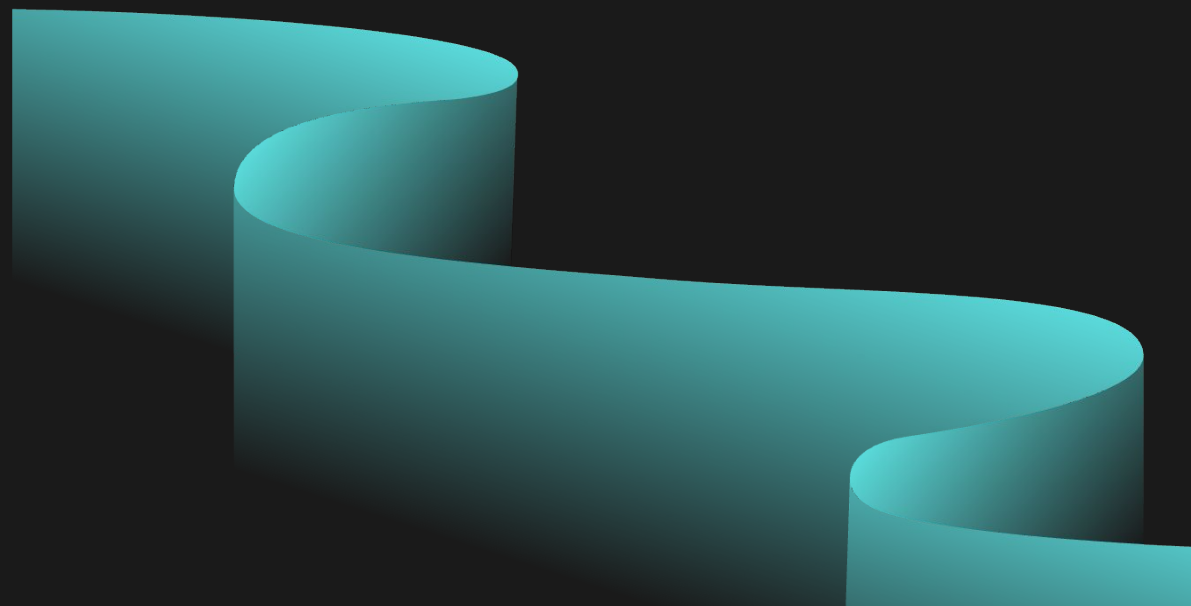
RESEARCH QUESTION 2

What title features most effectively drive audience engagement as measured by article views?

METHODS:

- 1) engineering new article title feature variables
- 2) LASSO regression for variable selection
- 3) OLS Linear Regression/ Variance Inflation

Factors (VIF)



New Article Title Feature Variables:

- **Title_Word_Count**
- **Title_Character_Count**
- **Avg_Word_Length**
- **Longest_Word_Length**
- **Title_Has_Verb**
- **Title_Has_I**
- **Title_Has_AI**
- **Title_Has_China**
- **Title_TFIDF_Max** - The highest TF-IDF score among all words in a given article title. This value represents the single most distinctive or unique word in the title relative to the rest of the corpus.

Lasso Regression Results:

<u>Variable</u>	<u>Coefficient</u>
Intercept	224.016
Title_Word_Count	0
Title_Character_Count	0
Avg_Word_Length	0
Longest_Word_Length	-11.084
Title_Has_Verb	59.095
Title_Has_I	10.359
Title_Has_AI	-26.528
Title_Has_China	48.317
Title_TFIDF_Max	12.819

OLS Linear Regression:

Table 8: OLS Regression Results: Article Title Features and Views				
Variable	Estimate	Std. Error	t-value	p-value
Intercept	210.155	65.919	3.188	0.0025 **
Longest_Word_Length	-12.144	5.489	-2.212	0.0316 *
Title_Has_Verb	70.336	27.716	2.538	0.0144 *
Title_Has_I	23.508	43.022	0.546	0.5873
Title_Has_AI	-37.018	26.603	-1.392	0.1704
Title_Has_China	62.289	27.994	2.225	0.0307 *
Title_TFIDF_Max	21.662	15.950	1.358	0.1806
Residual Standard Error: 79.4 (df = 49)				
Multiple R^2 : 0.326 Adjusted R^2 : 0.243				
F-statistic: 3.942 (6, 49), p = 0.0027				

Variance Inflation Factors:

<u>Variable</u>	<u>VIF</u>
Longest_Word_Length	1.285
Title_Has_Verb	1.567
Title_Has_I	1.091
Title_Has_AI	1.233
Title_Has_China	1.632
Title_TFIDF_Max	1.814

KEY TAKEAWAYS

What title features most effectively drive audience engagement as measured by article views?

✱ Title_Has_Verb - binary indicator that measures whether the title uses action-oriented framing

✱ Title_Has_China - binary indicator that measures whether the title contains 'China' or 'Chinese'

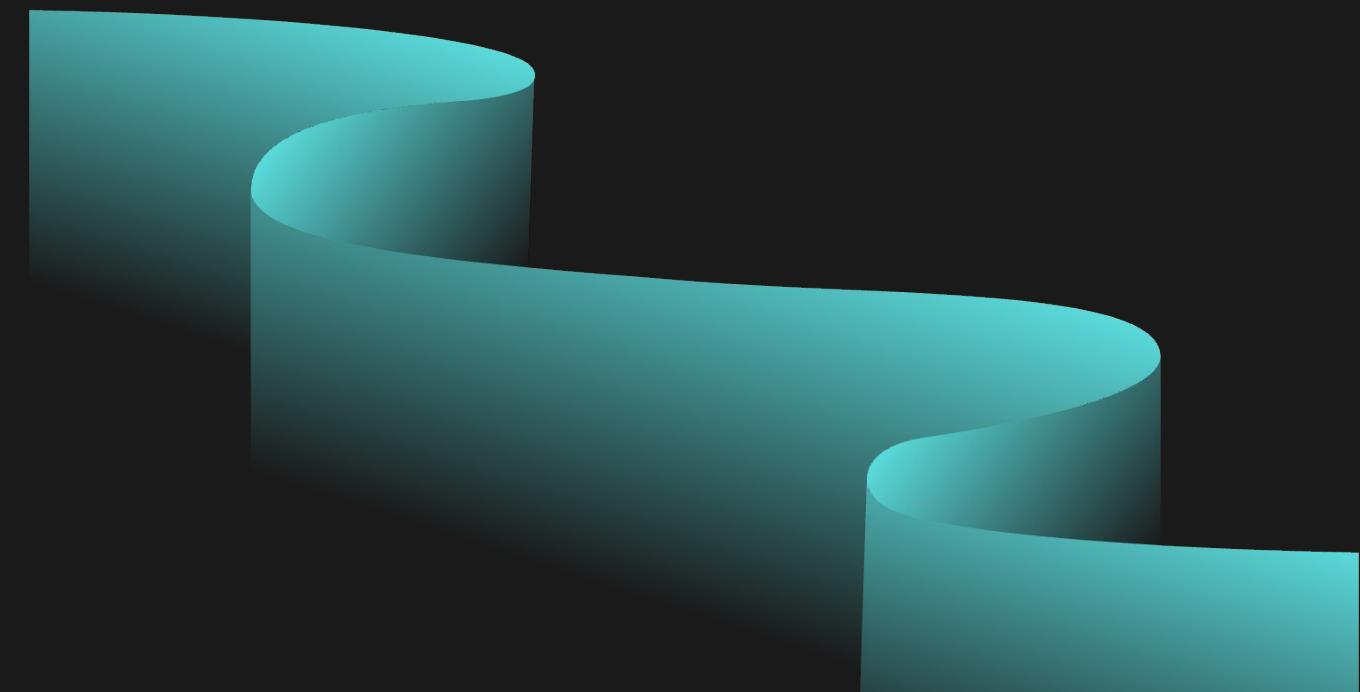
✱ Longest_Word_Length - number of characters in the longest word in the title

RESEARCH QUESTION 3

Which publishing day and hour are associated with higher article views?

METHODS:

- 1) reconstruction of time variable
- 2) Bootstrap Sampling for views by Publish Day
- 3) Bootstrap Sampling for views by Publish Hour



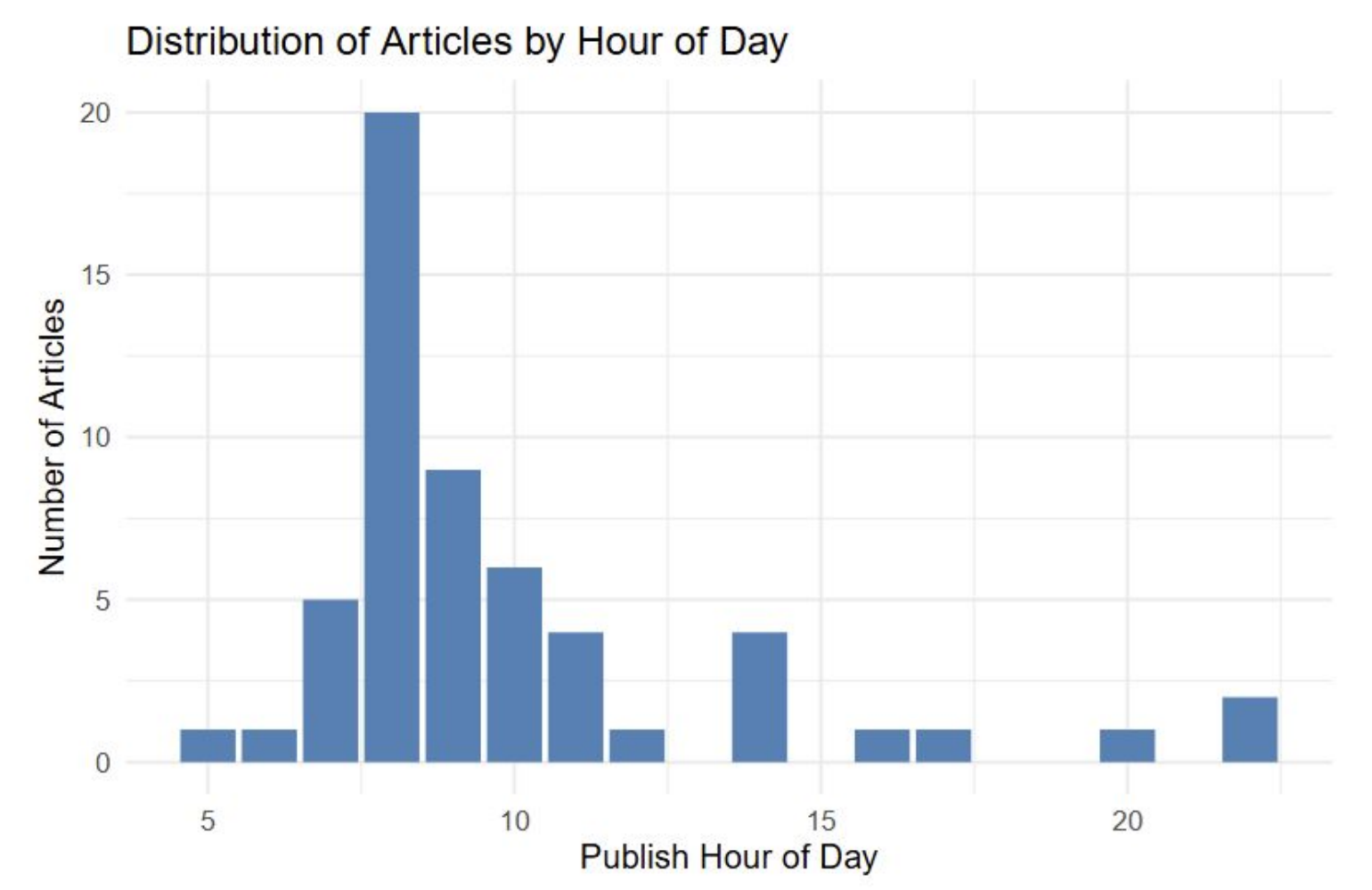
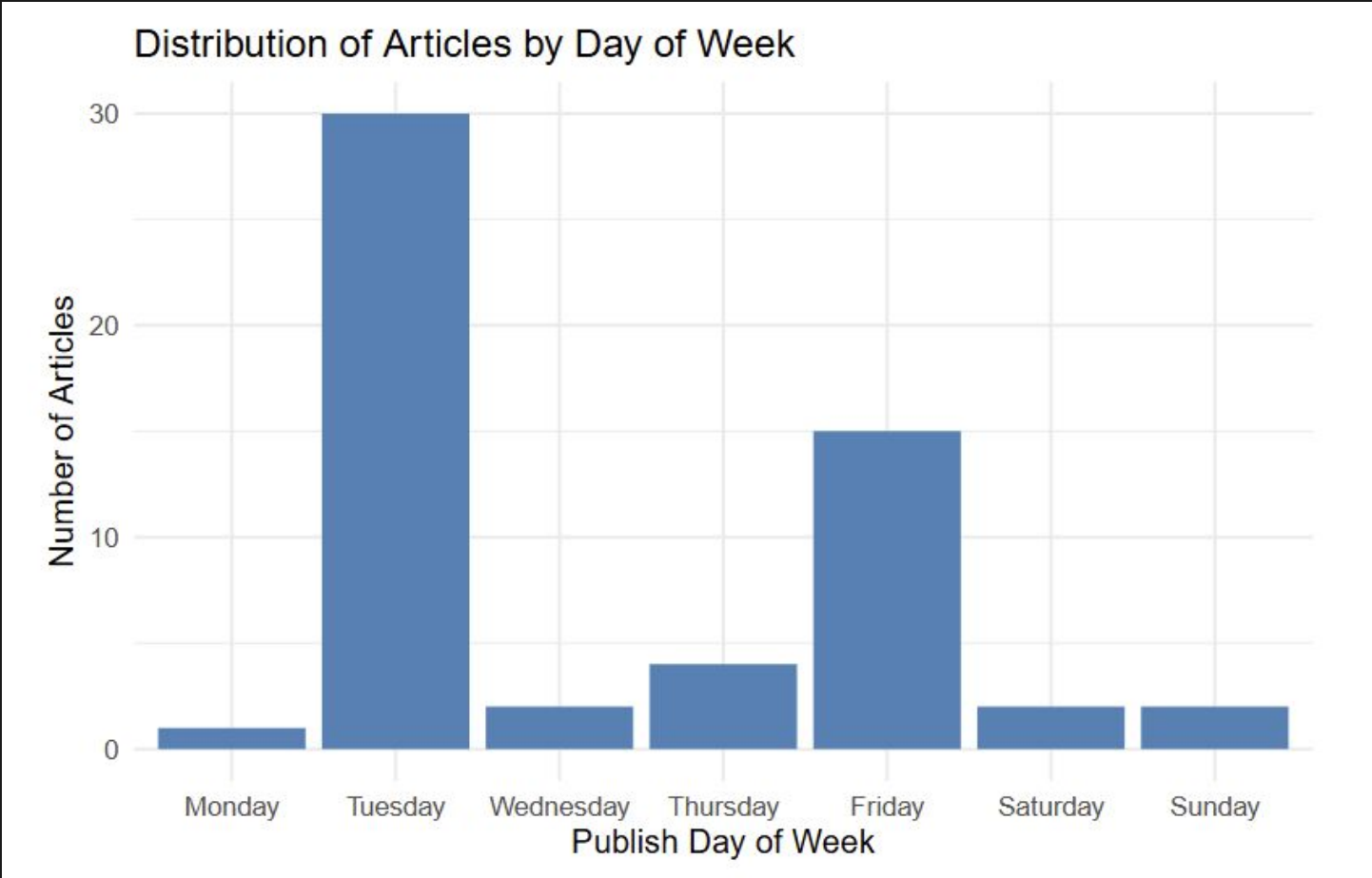
Reconstruction of the time variable:

The original time variable was recorded in clock format (eg. 8:55 AM).

We rounded each observation down to the nearest hour and converting it into a categorical variable taking values from 1 to 24.

<u>Time</u>		<u>Publish Hour:</u>
8:55 AM	➡➡	8
12:05 PM		12
7:30 PM		19

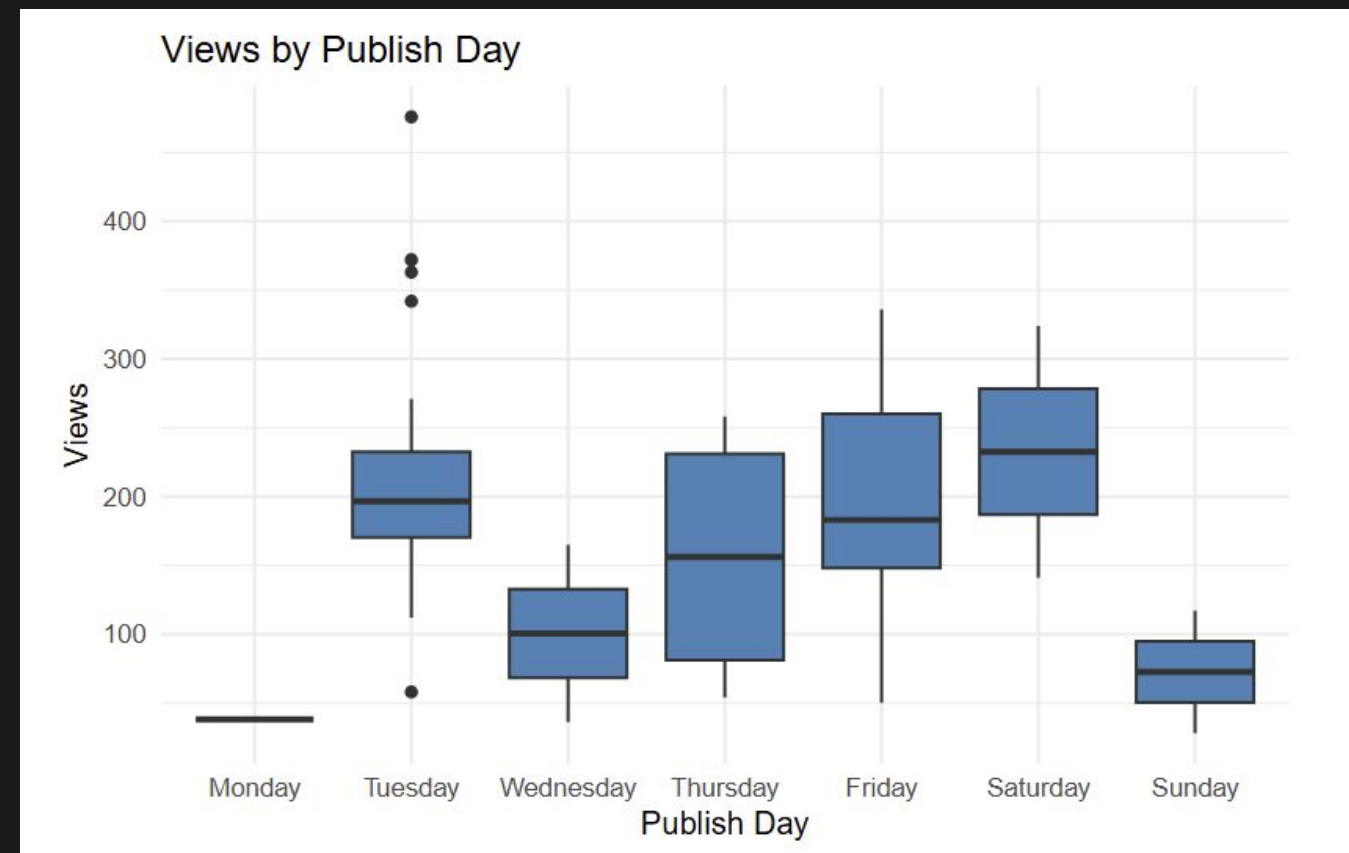
RESEARCH QUESTION 3



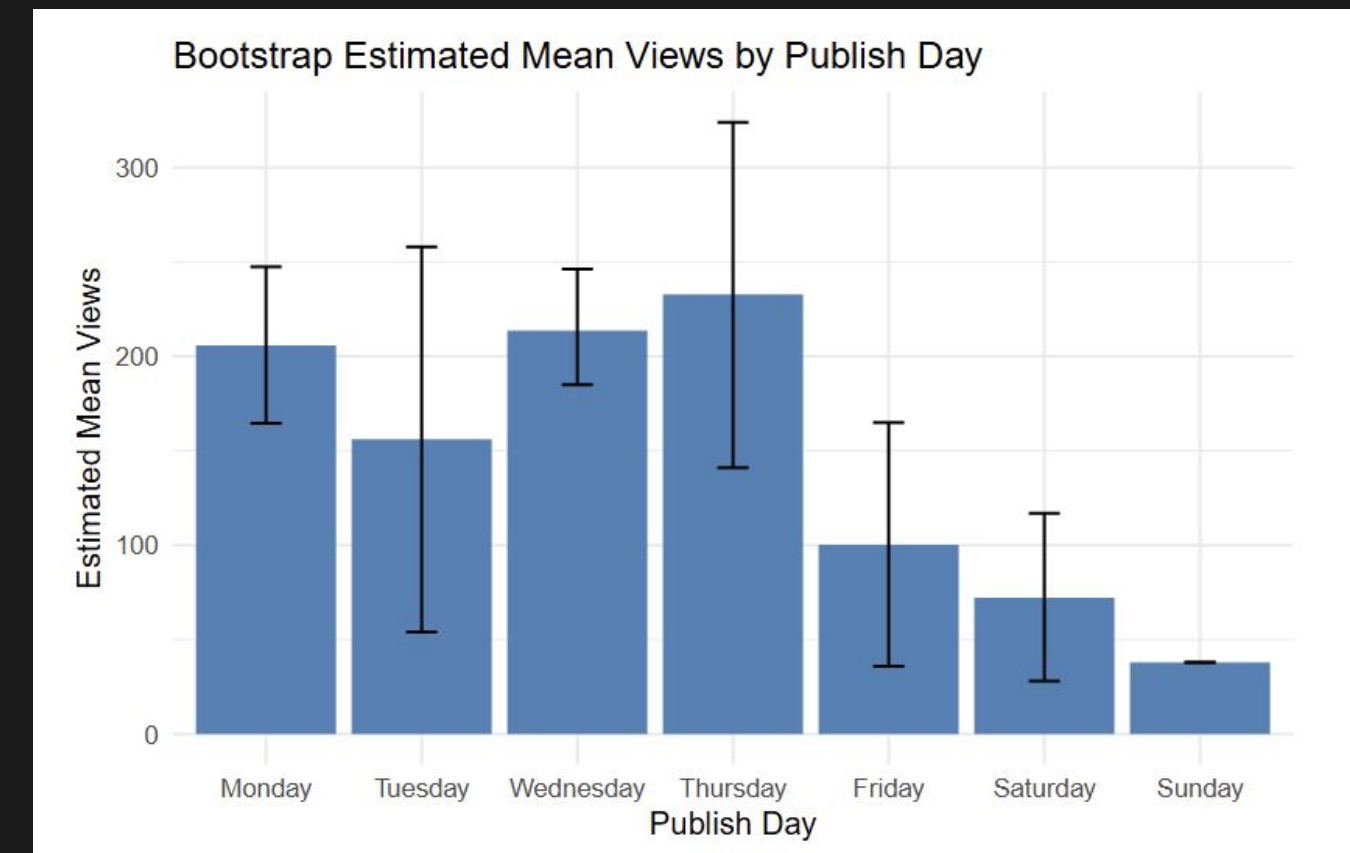
RESEARCH QUESTION 3

Which Publishing Days = Higher Views?

Observed Means From the Data:



Means After Bootstrap Sampling:

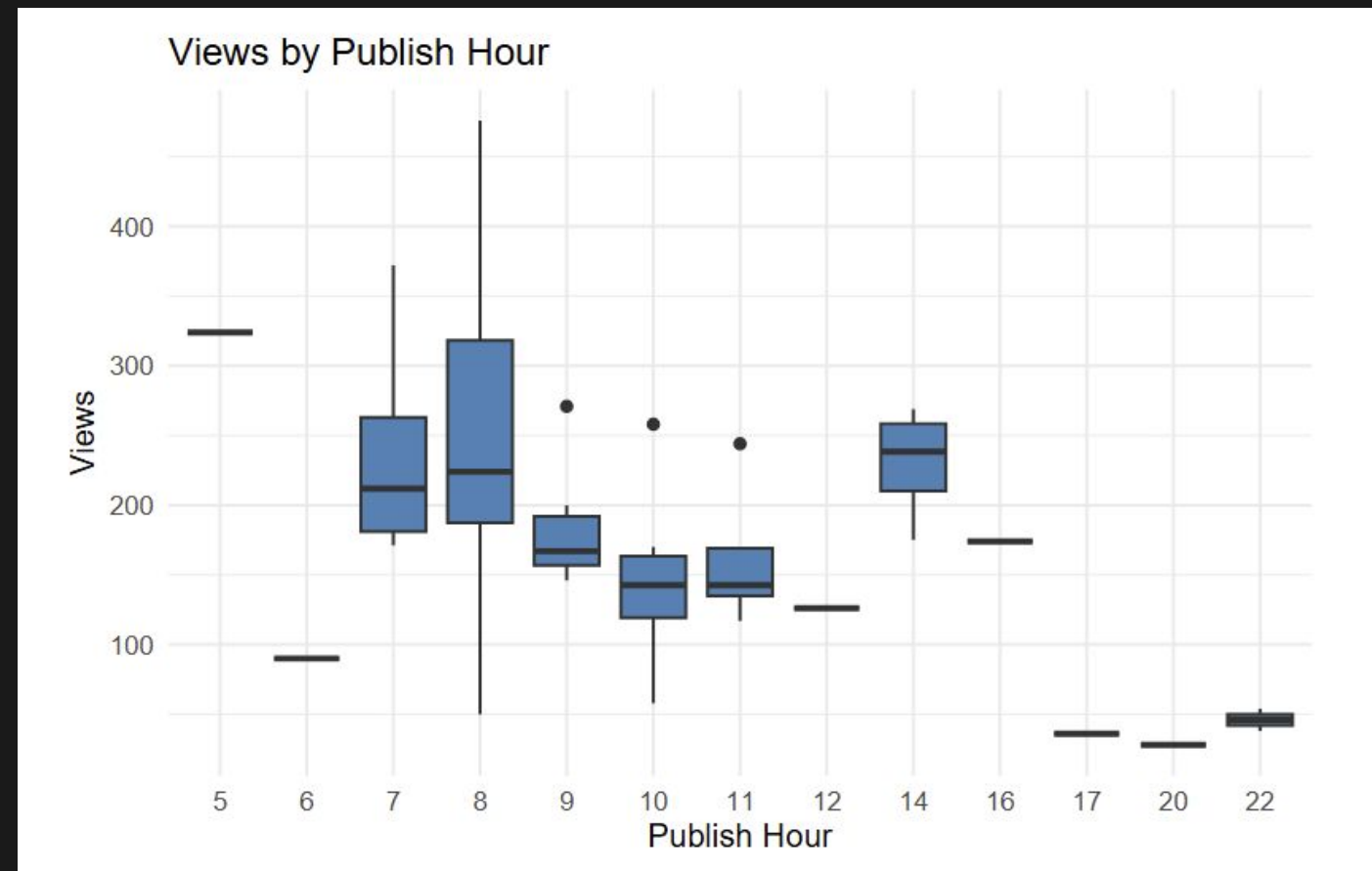


Bootstrap shows that Monday-Thursday show higher views (confidence intervals overlap).

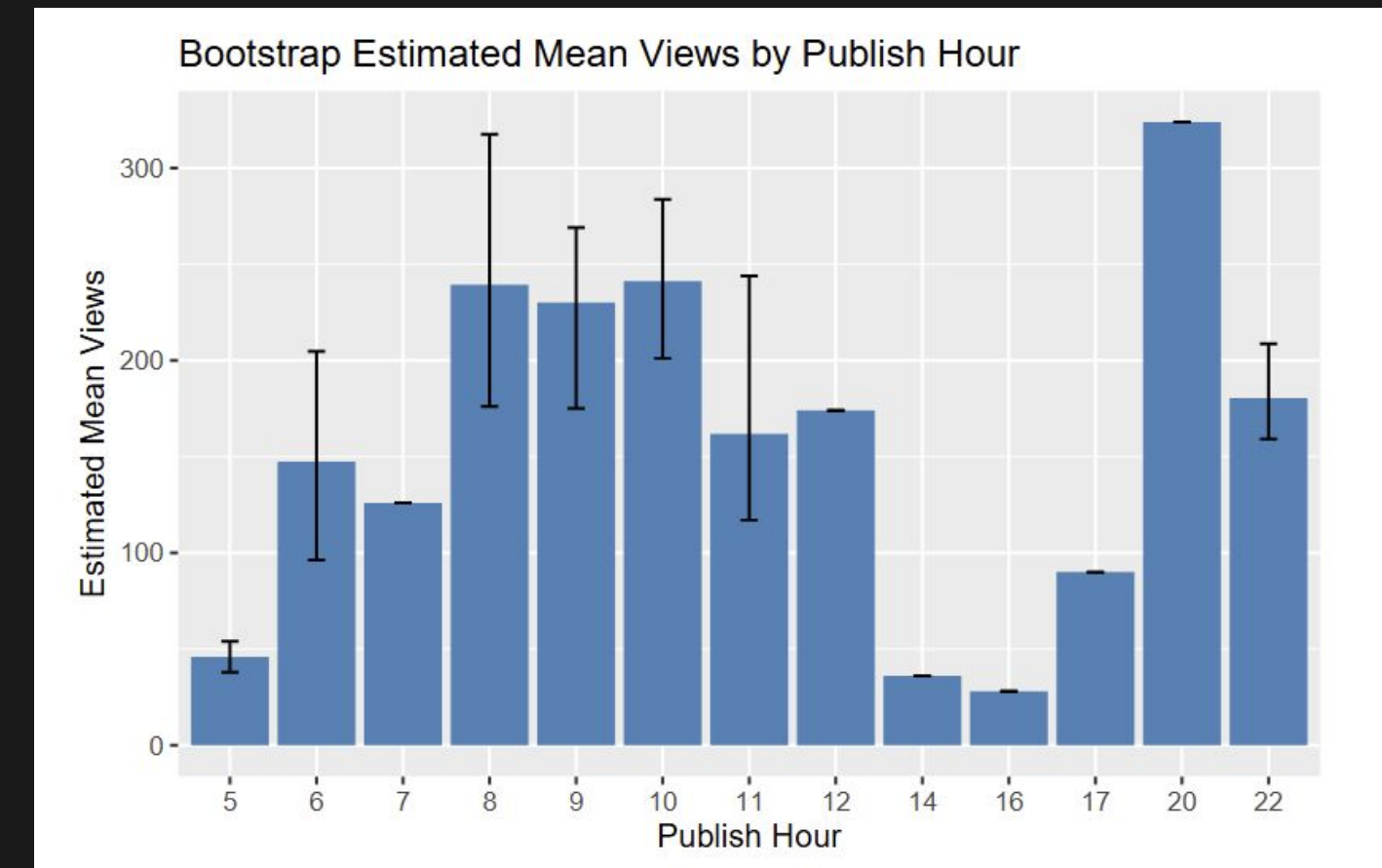
RESEARCH QUESTION 3

Which Publishing Hours = Higher Views?

Observed Means From the Data:



Means After Bootstrap Sampling:



Bootstrap shows that around 8-11 AM shows highest views.

KEY TAKEAWAYS

Which publishing day and hour are associated with higher article views?

* Monday–Thursday show higher views compared to Friday–Sunday.

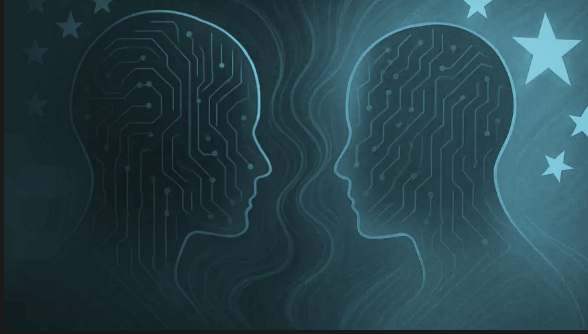


Weekdays > Weekends

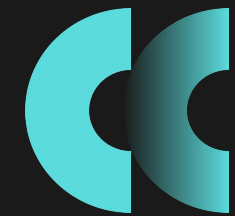
* 8–11 AM show higher views compared to earlier or later times.



Morning > Early Morning/Evening

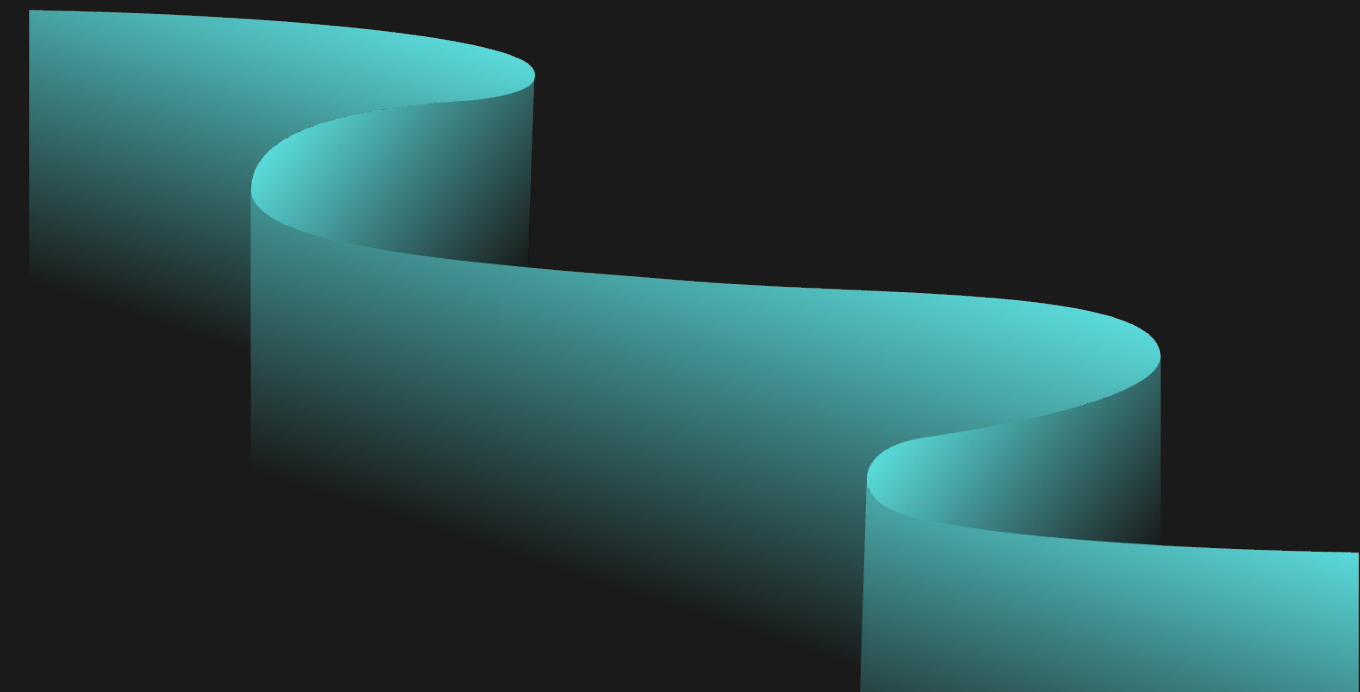


CONCLUSION/SUGGESTIONS



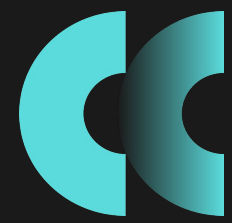
Article Titles

- Action-oriented titles and clear wording are associated with higher article views.
- Titles referencing China also tend to receive higher engagement.
- Titles with simpler wording tend to perform better than more complex wording.



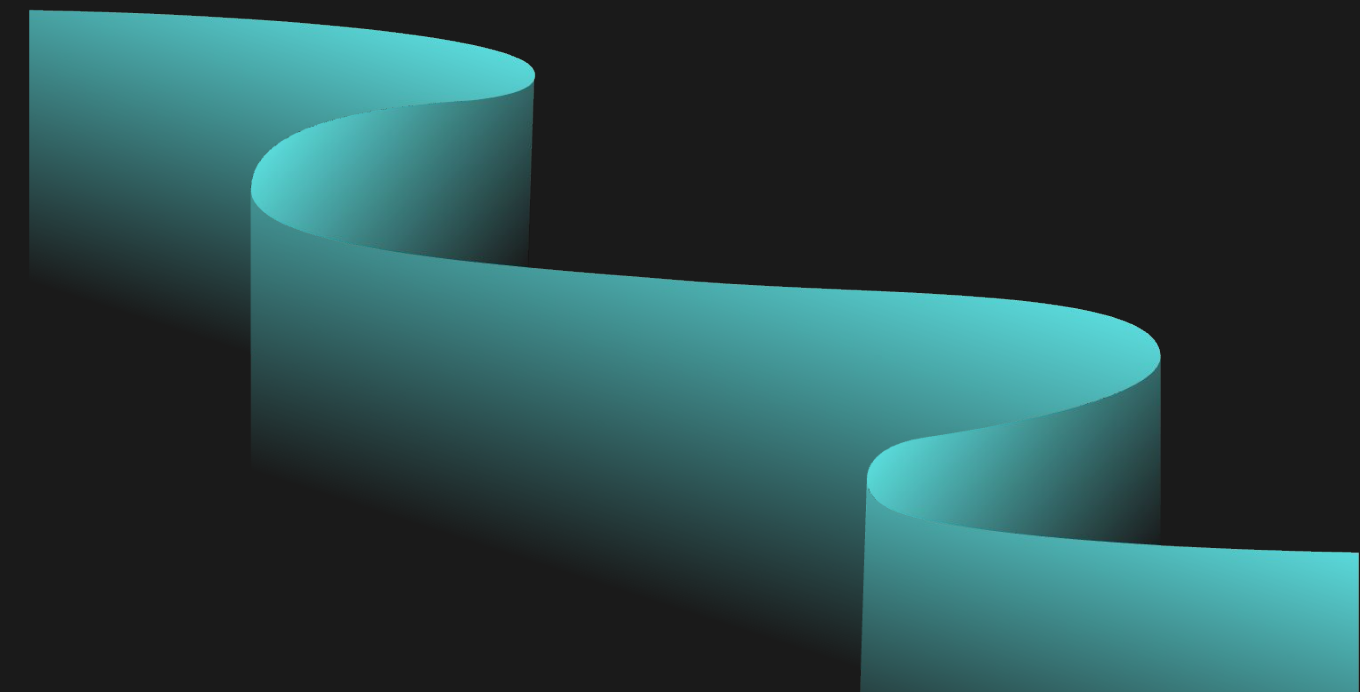


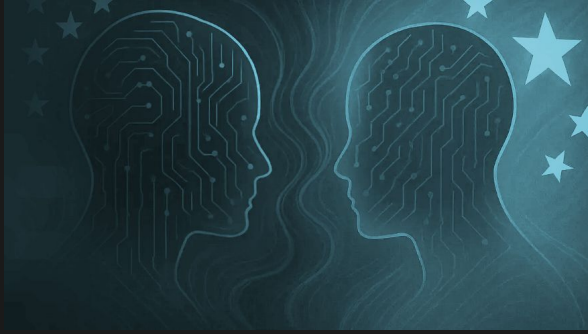
CONCLUSION/SUGGESTIONS



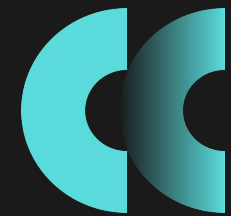
Publishing Time

- Most articles were published on **Tuesday** and **Friday**.
- Results suggest that 8-11 Am on **Weekdays** may be an effective publishing time.





CONCLUSION/SUGGESTIONS



Audience Growth Strategy

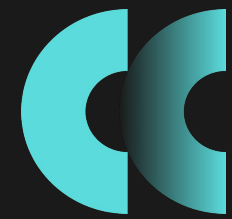
- The **Substack Network** brings highly engaged readers.
- Growth strategies may include:
 - collaborating with large Substack creators
 - increasing activity on **Substack Notes**



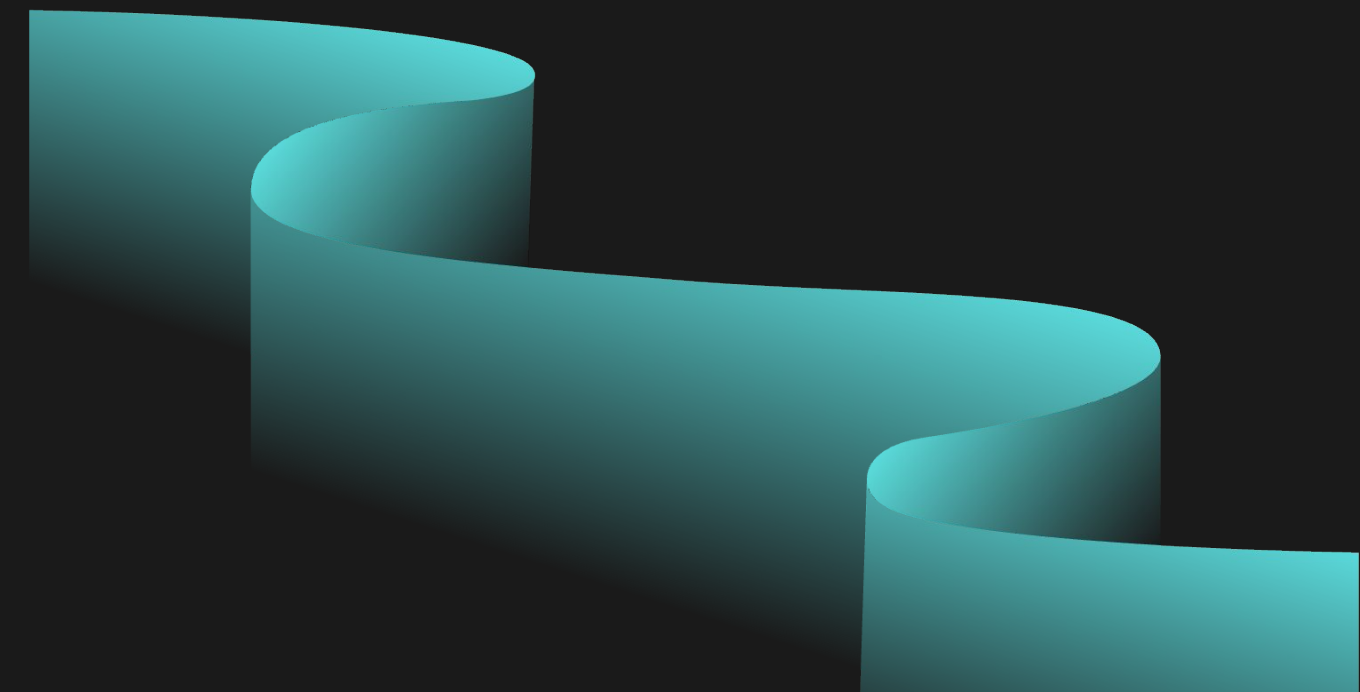
Overall, these findings provide practical recommendations for increasing readership and engagement.



Limitations



- Relatively **small dataset**, especially for article analysis.
- **Uneven publishing times** across days and hours.
- Data comes from **one newsletter**, so findings may not generalize.
- Analysis identifies **statistical associations** between variables but cannot establish causal relationships.



Thank you for your attention!

